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*** START OF THE PROJECT GUTENBERG EBOOK ESCAPE
VELOCITY ***

ESCAPE VELOCITY

BY CHARLES L. FONTENAY

It was a duel to the death and Kraag had all the advantages, including offense and defense. Jonner had neither, but he employed an old equation peculiarly adaptable to the situation. And the proper equation properly worked....

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Murdering Stein was easy. Kraag waited until Jonner donned his spacesuit and went out to have a personal look at the asteroid. Even then Kraag held his patience, because he wanted Jonner to come back to the ship unsuspecting.

Kraag sat tensely at the back of the control room while Stein, the navigator and communications man, operated the radio. There was a brief period when Stein talked with Marsport, then he got in touch with Jonner. Until Jonner got some distance from the wrecked ship, most of their conversation was an argument.

"I still think two of us ought to go out and one stay at the ship," argued Stein. "Kraag agrees with me. What if you fall into a crevice?"

"There's not much danger, and you've got a directional fix on me," replied Jonner's voice through the loudspeaker. "If we had a large crew, I'd agree we ought to explore in pairs. Since there are just three of us, only one ought to be endangered at a time. I'm the captain, so I'm it."

"Well, don't get out of sight," warned Stein. "We don't have an atmosphere here to bounce radio waves over the horizon."

Through the glassite port, Kraag could see Jonner poking around at the asteroid's surface with his steel probe. Against the incredibly curved horizon, Jonner's suited figure leaned at a slight angle under the black, star-studded sky. The distant sun gleamed from the sphere of his helmet.

"Pretty smooth terrain," remarked Jonner. "It's not much of a planet, but it seems to have enough mass to pull down any mountains. Looks like there should be some hills, though. It must have been in a molten state when the original trans-Martian planet was broken up."

"That ought to mean high albedo," said Stein. "Higher than it ought to be."

"Sounds more like Vesta," said Jonner. "Sure we're on Ceres?"

Stein looked at the notes he had made from the ship's instruments, before the crash.

"The escape velocity was 1,552.41 feet per second," he said, "and the diameter 0.06. I figure the mass at .000108."

"All those figures are off according to the latest table for Ceres," said Jonner.

"The fellows that made that table were on Mars," reminded Stein. "Vesta doesn't have a 480-mile diameter. It must be Ceres."

"You're the navigator," surrendered Jonner. "I'll take your word for it."

The personnel sphere of the ship rested on the ground, tilted at almost a 20-degree angle from the horizontal. The tilt was no inconvenience, however. Each of the men weighed only five or six pounds here, and slippage was hardly noticeable.

"I'll turn you over to Kraag," said Stein at last, glancing up at the chronometer. "It's my day to fix supper, you know."

It was the signal Kraag had been waiting for. He reached behind him and fumbled in the rack for a gun.

The one he brought out was Jonner's, and it wasn't a heat-gun but the ancient pistol Jonner swore by. Kraag put it back hurriedly, but not before Stein had turned in his chair and seen it.

"What's up, Kraag?" asked Stein without alarm. "Why the gun?"

Kraag pulled a heat-gun from the rack.

"Nothing's up," he said, and shot Stein.

The ray burned into Stein's shoulder, and Kraag swung it down across Stein's chest to his stomach before relaxing his pressure on the trigger.

"My God, Kraag!" gurgled Stein. Summoning a last effort, he croaked into the microphone: "Jonner! Watch out! Kraag shot...."

Kraag blasted him in the face, cutting him off. Stein's body floated forward and upward out of the chair and began to settle slowly toward the slanting floor.

"What's going on, Stein?" came Jonner's alarmed voice over the loudspeaker. "Stein? Stein!"

"It's all right, Jonner," said Kraag as calmly as he could, when he could reach the microphone. "Stein just fainted."

There was silence from Jonner.

"I'll take care of Stein and then take over the mike till you get ready to come in," said Kraag into the microphone.

"I want to talk to Stein when he comes around," said Jonner. His voice sounded cold.

So Jonner suspected something. Well, that couldn't be helped. Maybe he could be talked around.

"All right, Jonner," agreed Kraag soothingly.

Stein's body had to be hidden from Jonner, just in case. Jonner got into the personnel sphere alive—something Kraag did not intend for him to do. When he had taken care of Jonner, he could dispose of both bodies before the rescue ship got there.

Dragging Stein's body was like towing someone through water. It floated through the air of the sphere at Kraag's tug, settling slowly. His only problem was getting good leverage for pushing. After some cogitation, he jammed the body into an empty food compartment two decks below the control room.

Back in the control room, Kraag looked out the port. Jonner was closer to the personnel sphere now, looking toward it but not moving.

Other portions of the ship, some jettisoned, some crumpled and broken apart by its crash, lay at varying distances from the personnel sphere. Some of the parts were scattered out of sight beyond the horizon, a mile away.

Kraag had not wanted to fool with the asteroid. There had been no question that they had to swing back off their original orbit toward Titan when the meteorite slashed open both of their hydrazine tanks. But Kraag's idea had been to stay in space and try to turn back toward Mars before the fuel gave out.

As the engineer, Kraag resented Jonner overruling him. Jonner had felt it safer to take an orbit around the asteroid and wait for rescue. But the fuel pumps had failed before they could adjust to the orbit. Kraag would never forget that helpless waiting as they circled and circled, spiraling downward to the inevitable crash.

He went back to the microphone.

"Okay, Jonner," he said. "What's going on out there now?"

"Where's Stein?" countered Jonner. "I want to talk to him."

"He's not feeling so good. Said he'd rather not try to get back up to the control room right now."

"Tell him to come to the mike anyhow. I don't want to talk to you till I talk to Stein."

"Stein can't talk, I tell you. If you don't want to talk to me, then are you ready to come in?"

"And get shot?" retorted Jonner.

So Jonner's suspicions were that definite. It was to be expected after the words Stein had been able to shout into the microphone. Jonner was nobody's dumbbell.

Kraag tired once more.

"That's a ridiculous idea, Jonner," he said. "I can't figure why you'd say such a thing."

"You shot Stein," said Jonner positively. "There's no use your denying it. I know you shot Stein, and I'll know it until Stein himself tells me it isn't so."

Kraag knew Jonner too well to try to keep up the pretense any longer. He tried another tack.

"Okay, so I shot Stein," he admitted. "That doesn't mean I'll shoot you. Come on in and talk it over. We can make a deal."

"If you shot Stein, why wouldn't you shoot me?" asked Jonner logically.

"There wasn't enough air for three. There is for two."

Jonner was silent for a moment.

"So that's why you did it," he said then. "Figured it pretty close, didn't you, Kraag?"

"I'm the guy who has to watch supplies on this boat. I checked the oxygen after the crash broke open those three compartments on the supply deck. There's 3800 pounds of oxygen left. It'll take about 22 months for the rescue ship to get here from Mars. At 2.8 pounds of oxygen a day, you and I can make it, but it would have lasted the three of us only 15 months."

Jonner cursed him for a full minute, not loudly but with such intensity that Kraag felt his face getting warm.

"You damn murderer!" finished Jonner. "You damn cold-blooded murderer!"

"Cut it out, Jonner," growled Kraag. "I can't understand you and Stein. What were you expecting to save us? A miracle?"

"I don't feel like talking about it now," said Jonner warily. "If you had only ... Hell, Kraag, we'd been together a long time. Even if all of us had thought we were going to die, I didn't think we'd kill each other off like animals."

"Self-preservation is the first law of nature," said Kraag cynically. "Better that two should die than three. Come on in, Jonner."

"That's self-preservation? No thanks, Kraag. You know I'll turn you in as a murderer when the rescue ship gets here. I have no hankering to walk up where you can burn me down."

"Okay, stay out there till your air gives out."

The airlock was not a comfortable place to spend one of the asteroid's seven-hour nights, but Kraag was afraid not to stand guard there with his heat-gun. He was afraid to sleep, too, for the airlock combination was virtually noiseless and Jonner could open it from the outside. Jonner was unarmed, but Kraag had no hankering for a hand-to-hand fight with the powerfully built captain inside the personnel sphere. Because the air would swish out of the lock instantly if Jonner opened it, Kraag had to wear a spacesuit.

He tried to talk to Jonner several times, but got no answer. Toward dawn, Kraag dozed off, only to be brought awake with a start by Jonner's voice in his earphones.

"Good morning, Kraag," said Jonner. There was iron in his voice. "Have a good night's sleep?"

"About as good as yours, I'd say," retorted Kraag, wishing he could get his hands inside his helmet to rub his eyes.

"I slept fine. Found me a good foxhole just beyond the horizon."

"Damn you, Jonner! Where are you now?"

"Go on and have breakfast, Kraag. I'm far enough away for you to see me. Take a look."

Kraag peered out of the uppermost airlock ports, one by one. They slanted at a bad angle, but through one of them he made out Jonner, standing half a mile away. Uncannily, as though he could see Kraag's helmet at the port, Jonner waved.

Kraag was afraid to take off the spacesuit now because the supply deck had no ports and Jonner could get to the ship in a hurry if he wanted to. He took off the helmet, though, and went up to the center deck. Hurriedly, he opened the cover of the port in the direction he had seen Jonner. Jonner was still in the same place, sitting down.

Kraag heated breakfast and ate it with an eye on the port. Jonner didn't move. Kraag felt better when he had eaten, and went up to the control room.

"Why don't you give it up and come on in, Jonner?" he asked. "The oxygen in that suit's not good for more than another 15 hours."

"That's where you're wrong, Kraag, and that's what's so tragic about your murdering Stein," said Jonner quietly. "You either forgot that we carried oxygen instead of nitric acid as the fuel oxidizer this trip or, being an engineer, you didn't think of it except as fuel."

"There's enough oxygen in the tanks scattered over the landscape to keep a dozen men alive until the rescue ship gets here. It's hard for me to get at, but I've already found I can manage it."

Kraag was profoundly shocked. For a moment the enormity of what he had done in killing Stein almost overwhelmed him. It had been completely unnecessary.

Then his self-reproach turned into a growing anger against Jonner. Jonner was always so reticent, always required his orders to be obeyed without explanation. During the whole argument about taking an orbit around the asteroid, during the whole time it had taken to spiral down to a crash, he had not told Kraag how he expected them to stay alive until they were rescued.

Kraag hadn't asked him, of course. Kraag had assumed Jonner was thinking in terms of his own figures.

"I'm sorry about Stein," said Kraag, and meant it. "But it can't be helped now, Jonner. There's enough air for both of us, if you'll keep your mouth shut when the rescue ship gets here."

"If I promised, I still wouldn't trust you and you wouldn't trust me. No, Kraag. The only way it'll work is for you to come out unarmed and let me go in and get the guns. Then I'll lock you in the control room till the rescue ship gets here."

"One of us is a fool, Jonner, and you seem to think it's me. I'm not going to burn for murder. I've got the whip hand. You may have oxygen, but you've got to have food and water, too."

Jonner laughed, without humor.

"I've got enough of that for three Earth days and I can last longer," he said. "Before that time, I'll come and get you, Kraag. Don't go to sleep!"

Kraag cursed and switched off the loudspeaker. But he kept an eye on Jonner through the glassite. Always, he had to watch Jonner—or stay on guard in the airlock.

If there were only some way to lock Jonner out! But the only real lock was on the control room, and a man couldn't live in the control room with an enemy below who could cut the water and oxygen lines.

Kraag would have to sleep some time. Jonner couldn't know when, but Jonner already was seven hours sleep up on him. Jonner could pick his own time to slip up to the sphere under cover of darkness, he could pick his own time to come through the lock. Maybe Kraag would be awake and could burn him down—but maybe not.

There was only one thing to do. He'd have to take the attack to Jonner.

Still watching Jonner through every port he passed, always watching Jonner, Kraag hung a heat-gun on one of the hooks at his spacesuit's belt. He went back below, put the helmet on, and went out through the airlock.

The shadow of the sphere stretched away toward his left. He was in sunlight. Jonner, still in the same spot, got to his feet but made no move to approach.

"Welcome to the great outdoors," said Jonner.

"I'm going to get you, Jonner," said Kraag grimly. "One way or another, I'm going to get you."

He moved toward Jonner. Each step was a long, floating leap and it was hard to stay balanced before landing. Jonner moved, not away from him but sidewise.

Kraag stopped. The effective range of the heat-gun was no more than 100 feet. If he tried to get close enough to Jonner to use it, Jonner could circle and get to the personnel sphere.

There were the oxygen tanks, the big ones used for fuel. If Kraag could get to them and burn them open, Jonner couldn't last long outside. But they were scattered pretty far from the personnel sphere. Jonner would get to the sphere for sure if he tried that.

"Okay, Jonner, I know when I'm licked," said Kraag. "Come on in."

"I'm not too far away to see the gun, Kraag."

"I'll take it back to the sphere and leave it."

"Why not just toss it away?"

"And have you beat me to it and get the drop on me? We'll leave the guns in the sphere and I'll meet you on even terms."

"I'll believe it when I see it."

Kraag went back to the sphere. He couldn't stand in shadow without looking suspicious, but he took the heat-gun from his belt ostentatiously and swung it in an arc, apparently tossing it through the open outer lock. Instead, he held onto it and hung it by the trigger guard to a belt hook at the back of his suit.

"I'm all clean, Jonner. Come on up," he invited.

"Let's see the hooks, Kraag," said Jonner.

Kraag held his arms aloft, wriggling the empty steel fingers of the spacesuit. Jonner came toward him, floating high above the surface with each step. At just about the extreme range of the heat-gun, he stopped. Kraag kept his arms outspread, but tensed himself.

"Clean, so far," said Jonner drily. "Now turn around, Kraag."

"And have you jump me from behind? Not hardly."

"Gun on the back hook, eh, Kraag?"

"Damn you, Jonner!" Kraag reached behind him for the gun and at the same time leaped toward Jonner. Jonner, ready, jumped back, and Jonner was a more powerful man. Handling a heat-gun with the hand-hooks of a spacesuit is awkward business, and by the time Kraag could bring the weapon to bear on Jonner and press the trigger, Jonner's distance was such that the ray obviously did no worse than make things uncomfortably warm for him.

"I didn't think that surrender rang true," commented Jonner. "If you'd been level, you'd have tossed away the heat-gun."

Then Jonner revealed that he was not entirely weaponless. As he hit the surface, his arm moved in an arc and a good-sized rock came hurtling through space toward Kraag.

Kraag writhed frantically, two feet off the ground, and the stone missed him by inches. Kraag landed on his side and bounced again. Jonner hit once more and hurled another rock. Evidently he was armed with several of them. This one ricocheted off the ground near Kraag just as Kraag finally slid to rest.

Getting to his feet and turning to flee was agonizingly slow, when every frantic movement lifted him off the ground. Another stone came sailing by, to strike the personnel sphere and rebound at an angle, before Kraag could jump back, away from Jonner.

Perspiring and panting, he clambered hastily back into the safety of the airlock.

Jonner's rocks were a better weapon than a heat-gun, Kraag realized. They weighed only a fraction of an ounce and Jonner could fling them an amazing distance. But their mass was just the same as ever, and a jagged one could rip a fatal hole in a spacesuit. He had no intention of engaging in a stone-throwing duel with Jonner, in which Jonner would be at least on equal terms with him.

On the other hand, it was even more imperative than before that he eliminate Jonner as soon as possible. A rock could be a deadly weapon if Jonner got inside the sphere, too.

At any rate, there was no point in concealing Stein's body from Jonner any longer and Kraag couldn't take chances on it polluting the atmosphere of the sphere. He dragged the corpse from the food compartment, down to the airlock, and pushed it out onto the surface of Ceres. The body settled stiffly to the ground a few feet away.

Kraag removed his helmet and hand-hooks, went back up to the control room and settled himself to watch Jonner. Jonner walked around freely, periodically hurling rocks at the sphere. The rocks bounced off without damage, but every time one of them hit the hull, the sound of it rang through the sphere.

Kraag switched on the communications system.

"Do you have to do that?" he demanded in irritation. "It's not doing you any good."

"Keeping me in practice," replied Jonner cheerfully. "I developed a pretty good arm throwing grenades in the Charax Uprising."

Jonner was a veteran of that brief but savage war on Mars, and sometimes reminisced about it. It was there he had developed his preference for the old-style projectile pistol over the heat-gun.

Kraag's eyes lingered on Jonner's pistol, hanging in the rack with the heat-guns, and slowly an idea spread through his mind. The heat-gun range was the same anywhere, but the range of a projectile weapon should be greater here than on Mars or Earth. Its range should be far greater than Jonner's rocks.

Kraag took it from the rack and turned it over in his hand, studying it. He wasn't sure of its principle, but thought it was something on the order of rocket fuel. It should fire without an atmosphere around it.

There were some figures stamped on the barrel: "COLT 1985, Cal-.45, MV-1100, Ser-45617298." Kraag puzzled over them. He knew the first one was the make and year and the last undoubtedly was the serial number. He deduced that "MV-1100" probably was a figure showing the relationship

between the projectile's mass and velocity. But it had been a long time since projectile weapons were common.

He called on the memory of a demonstration of the weapon Jonner had given his companions once on Mars. There was something that had to be done to prepare it for firing. Holding it in his right hand, Kraag grasped the barrel with his left. After a moment of hesitant tugging, he hit the right movement and the whole outer casing of the barrel slid backward and clicked. It snapped back into position as Kraag released it, and he remembered.

The gun was primed now. All he had to do was press the trigger and it would fire. It would automatically prime itself again after firing. It would fire each time he pressed the trigger now, until it exhausted its projectiles.

Exultant, he laid it carefully in a contour chair, where it wouldn't slide out. He put his helmet back on and replaced the hand-hooks of his spacesuit.

He looked out several ports before he found Jonner. The captain was not more than 150 feet away, casually lobbing rocks at the sphere.

Kraag picked up Jonner's pistol and made his way down to the airlock. He emerged and walked around the sphere to the side where he had located Jonner.

Jonner was moving away now, though he couldn't have known Kraag was coming out. He was about 300 feet away—too far for a heat-gun, but certainly within range of the projectile weapon. He seemed to be headed toward one of the big fuel tanks.

Kraag levelled the pistol toward Jonner and pulled the trigger. To his astonishment, he was hurtled backward, heels over head.

The kick of a .45 on an asteroid is pretty powerful. Kraag must have bounced 50 feet backward over the terrain before he slid to rest on his stomach. But he held on to the pistol—and, since he never had a chance to release the pressure of his hand-hook on the trigger, it did not fire again.

When he struggled upright, Jonner was standing at the edge of the fuel tank, watching him.

"Using my gun now, eh, Kraag?" Jonner said. "You'd better stick to weapons you know something about."

With that, he disappeared behind the fuel tank.

Kraag got to his feet and advanced confidently. His heat-gun was still hanging at his belt if he got close enough to Jonner to use it, and he could fire the projectile weapon at Jonner when Jonner was out of heat-gun range.

He was learning. One had to point the projectile gun accurately before firing. It couldn't be swung around and focused while pressing the trigger, like a heat-gun. He might miss a few times, but he ought to be able to hit Jonner at least once before the ammunition was exhausted. Once should be enough.

Heat-gun ready in his left hand, projectile gun in his right, Kraag circled the fuel tank. Keeping it between them, Jonner had headed straight for the horizon, running in long, shallow leaps. He was at least half a mile away.

Kraag pointed the projectile pistol and pulled the trigger. Nothing happened. Then he realized that he had never released the pressure of his hand-hook on the trigger after firing the first time. He let up on it and pressed it down.

And again Kraag was hurled backward, but this time he was smashed against the fuel tank and rebounded forward, falling on his face. By the time he reached his feet again, Jonner had vanished over the horizon.

Cursing softly, Kraag made his way back to the personnel sphere. He had hoped to get Jonner with that shot. He was very sleepy, and now he was faced with another night on guard.

He entered the airlock, pushed himself gently upward to catch the rungs of the metal ladder and turned the wheel of the airlock's inner door.

Nothing happened. The door did not open.

Fear gripped him like a paralyzing hand. For a moment he thought Jonner had managed to get to the sphere ahead of him and somehow had locked him out. But that was impossible. Then he thought the inner door might be jammed, and he and Jonner locked out together.

He glanced frantically below him, then broke into relieved laughter. He had left the outer airlock door open. As a safety measure against the sphere's accidentally losing its air, neither door would open unless the other was shut.

And that meant he could lock Jonner out of the sphere simply by leaving the inner door of the airlock open!

His laugh was full and genuine now as he pulled the outer door closed.

"Having fun, Kraag?" asked Jonner in his earphones.

"Just looking forward to a good night's sleep, for a change," retorted Kraag triumphantly. "Prowl around all you want to, Jonner. I can wait you out, now."

"The airlock, eh? I wondered when your guilty conscience would settle down and let you remember about that airlock," said Jonner phlegmatically. "You know, Kraag, I had no idea you wouldn't think about a simple thing like that, till I looked through the airlock port last night and saw you huddled up there with a heat-gun. You should have turned out the light."

Jonner was silent for a few minutes. Then he added:

"I don't think I'd laugh yet, though, Kraag. Remember, you're fighting with my weapons."

Kraag wasn't sure what he meant by that: whether he was talking about Kraag's using the projectile pistol or the fact that they were in space, Jonner's natural element. Kraag himself had been in space 10 years, most of it with Jonner, but before then he had never left Earth. Jonner had been born and raised on Mars, where a man needed a suit to go to the next settlement, and he had been on a ship since he was 15.

As for using the pistol, Kraag could see danger for no one but Jonner. He had proved, twice, that he could fire it. He was quite sure the old-fashioned weapon was no more likely to explode than a heat-gun. The only trouble he foresaw was figuring how to reload it if he used up all its projectiles before hitting Jonner.

Kraag shrugged and removed his suit. He was hungry, and he was looking forward to a supper better than Jonner had available in the concentrated supplies in his spacesuit. Jonner's food and water by now had dwindled to less than 60 hours' supply, unless he was weakening himself by going on slim rations.

As he wolfed down his supper, Kraag took stock of his situation. He could see no flaw in his position. All he had to do was sit back and wait.

He decided not to destroy the tanks that were Jonner's supply of extra oxygen. After all, Jonner could not last beyond his food and water supply. The presence of the oxygen made his case airtight. He could dispose of the bodies of Stein and Jonner and tell the crew of the rescue ship they had wandered off on an exploration tour and never returned. With plenty of oxygen for the three of them, no motive could be established against him for the murders.

He began to feel rather sorry for Jonner. They had been companions, and Stein with them, for a long time.

After eating, he went up to the control room and turned Jonner in on the communications system. He was genuinely regretful that Jonner had to die so soon. It would be lonesome on the asteroid with no one to talk to.

"I hope you've been keeping the radio open to Marsport, in case there were any inquiries," said Jonner. "If they get the idea we're all dead out here, they may call off the rescue."

"The last time they called was right after you left the ship," said Kraag. "Stein was going to tell you, but I suppose he forgot it. Marsport knows where we are. A rescue ship should have blasted off by now."

"That's the advantage of being on Ceres instead of in space," Jonner pointed out. "They know Ceres' orbit, but they'd have to have several directional fixes on us, spaced several days apart, to pinpoint us if the ship were in space. What did Stein say the escape velocity here is?"

Surprised at the unexpected question, Kraag consulted the notes Stein had left lying in the control room.

"EV 1,552.41 feet per second," he replied. "Not figuring on jumping off the planet, are you, Jonner?"

"Maybe," said Jonner.

"Well, don't wake me up if you do. I'm really going to pound the pillow tonight."

Jonner laughed shortly, and Kraag heard the click as the captain switched off his helmet radio. He grinned.

Kraag was asleep almost as soon as he hit the bunk.

He came awake slowly, reluctantly, knowing he had not had all the sleep he needed. Something was pounding noisily somewhere, ringing through his head.

He shook his head to clear it. For just an instant there was silence in the utter darkness. Then:

CRASH!

Like a clap of thunder the noise reverberated through the metal hull of the sphere.

Kraag started violently, and only the bunk straps kept him from rocketing to the ceiling. Again:

CRASH!

And Kraag could feel the sphere shiver with the blow.

He switched on the lights just as another terrific crash sounded. This time he could see everything on the central deck quiver with the impact.

One of the four small ports around the central deck was uncovered, and the light threw a beam out into the black night of the asteroid. It brought a temporary cessation of the regular blows. During the interval, Kraag unstrapped himself and tumbled up to the control room, to switch on the communications system.

"Jonner!" he shouted. "Jonner, what in hell?"

"I'm not deaf," said the loudspeaker resentfully. "Give me a chance to turn down my volume, if you're going to holler."

"What the devil are you doing out there, Jonner?"

"What I promised you. I'm coming in after you."

Kraag swore.

"I'm going to blow you off the damned planet," he threatened, and leaped for the gun rack.

"You'll have to come outside to do it," reminded Jonner. "If you try to shoot through the ports, you'll save me a lot of work."

Kraag raced up and down the sphere twice before he had sense enough to turn out all the lights and use the searchlight. Then he located Jonner, clinging to the sphere outside the astrodome on the navigation deck. Jonner had a sledge hammer from the ship's cargo section in his hand.

Jonner grinned at him and moved quickly out of the searchlight's beam. Ten seconds later, another thunderous crash sounded, apparently from the other side of the sphere. Kraag swung the light in a circle, but Jonner could move faster than the beam.

Hastily, Kraag made another tour of the sphere, this time closing all the metal covers over the ports. When he reached the control room, Jonner's voice was calling him over the loudspeaker, repeating his name every few seconds.

"What do you want?" demanded Kraag, panting.

"Just wanted to tell you I could have knocked out the astrodome or one of the ports before you woke up," said Jonner cheerfully. "I don't want to kill you, Kraag. I just want you to surrender, and if you don't I can eventually batter through the meteor shield and the hull, and ruin the sphere for you."

"We'll see about that," gritted Kraag. Hurriedly he donned a spacesuit. Hanging Jonner's pistol at his belt, he took a heat-gun in his right hand and a flashlight in his left and ventured out through the airlock. He did not make the mistake of switching on the airlock light, but Jonner seemed to know when he emerged, possibly from the vibration when the lock opened.

"Nice night out, isn't it, Kraag?" Jonner welcomed him.

Kraag grunted. The night was black as pitch. The only way he could tell where the ground ended and the sky began was that the sky was jewelled with stars.

He turned the light on and flashed it over the sphere. No sign of Jonner. But a rock struck his helmet and bounced off with a clang that nearly knocked him down and left him momentarily dizzy.

"I'm behind you, Kraag," said Jonner pleasantly. "Better go back inside. I promise not to break your shell open tonight."

Kraag twisted around and fired the heat-gun even as he searched for Jonner with the flashlight. Both beams pierced emptiness. Jonner just laughed at

him.

Afraid now that Jonner would get into the sphere, Kraag scuttled back around to the airlock. Heat-gun ready, he turned on the light before closing the outer door, and breathed a sigh of relief at finding it empty.

Trembling with reaction, he closed the outer lock, left the inner one open and made his way up to the center deck. He needed coffee.

"I see you've gone back to the heat-gun," said Jonner. "That's smart."

"You'd like to see me exhaust the fuel tank of your pistol shooting it in the dark, when I can't hit you, wouldn't you?" retorted Kraag. "No, thanks. I'll keep it for long distances."

"Fuel tank? Oh, you mean the magazine." Jonner laughed. "I'd stay away from that old .45 of mine if I were you, Kraag. It's been with me too long. It's a lot more likely to turn on my enemies than to do me any harm."

"Rot!" snapped Kraag. "It's a gun. All I have to do is get the hang of aiming it properly."

"I wouldn't use too much power tonight, either," warned Jonner. "You don't get much with the solar mirror this far out. Anyhow, I took the mirror off while you were having your nap. The batteries should give out in a few hours."

Without answering, Kraag switched off his radio and removed his helmet. That last bit of information was a blow. Gradually, Jonner was stripping Kraag down to his own subsistence level.

Power or not, Kraag was determined to have his coffee. But first he went over the sphere again and switched off all unnecessary lights.

Jonner was a man who kept his word, but Kraag couldn't afford to trust him. Jonner might change his mind and try to break open the sphere again before morning. Kraag kept his spacesuit on. He did not sleep too well, for about once every 30 or 40 minutes something—either a large rock or Jonner's sledge hammer—would strike the sphere a resounding blow.

When Kraag's watch told him it was morning, he opened the ports of the center deck and let the weak sunlight stream into the sphere. Off to the east, he saw Jonner digging with a pick from the cargo. Jonner was far enough away for his legs from the knees down to be hidden by the extreme curvature of the little planet.

Kraag's first impulse was to go out and take a pot shot at him. Instead, he switched on the short-wave cooker and prepared some breakfast. Taking it up to the control room, where he could switch on the communications system, he opened the eastern port and watched Jonner. This high, he could see Jonner's feet and the hole he was digging—and Stein's body.

Jonner had taken Stein's body from the spot outside the sphere where Kraag had pushed it. He was burying Stein.

Jonner finished his excavation and laid Stein gently to rest in it. He pushed rocks back in to fill it up, and wrested a boulder that would have weighed a ton over it for a monument. Then he murmured a brief prayer over the grave.

Kraag was ashamed and then, unaccountably, angry. But he stood at the port, drinking his coffee and watching Jonner, and said nothing.

Either with chalk or with some soft rock he had found—Kraag could not tell which—Jonner wrote something on the big stone that was Stein's monument. Then he stood up and turned toward the sphere.

"Kraag," he said. "Kraag, are you tuned in?"

"Yes," replied Kraag shortly.

"You have today to surrender. Tonight I'm going to hatch you out of your comfortable egg."

Kraag switched off the communications system and paced the room, anger burning slowly inside him. This was ridiculous. He held all the cards. He had the guns, he had the sphere. Jonner was outside, weaponless, with a limited supply of food and water. Yet Jonner had him on the defensive.

How had it happened? How could it happen? Kraag lit a cigarette and puffed at it slowly, applying his mind coldly to the situation.

He didn't doubt that Jonner would do as he threatened, but he didn't think it was the recklessness of desperation. More likely, Jonner deliberately,

calculatingly, planned to reduce his own chances for comfort, in order to bring Kraag down to more even terms with him.

If Jonner broke the hull of the sphere, it could be repaired—by someone working outside, free from interruption by an enemy. Until it was repaired, it would mean that Kraag, too, would have to live in a spacesuit. And Jonner might knock open a hole, or more than one, big enough to permit him to enter the sphere and attack Kraag in the darkness.

If only he could surround the sphere with light at night, he could keep Jonner at a distance. But with the solar mirror gone, the searchlight, on top of the sphere's other electrical requirements, would discharge the batteries before the night was half gone.

Kraag knew Jonner's stubbornness, his resourcefulness, his raw courage. Jonner was the one of them who was really at bay, when you considered it. Yet Kraag felt that Jonner was closing in on him, gradually, inexorably.

Facing this, Kraag felt the steel enter his own will. He wasn't a coward. He had just been expecting this to be too easy. If Jonner would force him to fight, he would fight. He still had the advantage.

He must abandon the sphere as an asset. Jonner could take that away from him anyhow. On the other hand, if Jonner took over the sphere, Kraag could use the same weapon against him. He could break open the sphere.

So the sphere was no longer a factor. The food and water were no longer a factor, for food and water went with the sphere. He would admit Jonner to equality in those supplies—not full equality, for he could provision himself now more fully than Jonner had been provisioned two Ceres days earlier. He still might pin Jonner down as Jonner tried to get to the sphere for more supplies.

Then Kraag's remaining advantage lay in the guns. They should be enough. If he could get close enough to use a heat-gun, he could blast Jonner. Jonner's own projectile weapon would keep Jonner out of rock-throwing range, and sooner or later he would hit Jonner with it. He couldn't keep on missing; the law of average would give him a hit sooner or later. And all he needed was just one....

Kraag provisioned his spacesuit and hung all three of the heat-guns at his belt. In one of the capacious outside pockets he put two spare flashlights and

half a dozen of the extra fuel packets—What was it Jonner had called them? Magazines, that was it—for Jonner's projectile pistol. He took that pistol in his right hand and sallied forth to do battle.

Jonner was nowhere in sight. Kraag shut the outer lock to make it appear he might be still in the sphere if Jonner happened not to spot him. He went over to Stein's new grave.



Jonner had written on the stone: REST IN PEACE. R. STEIN MURDERED BY A. KRAAG. DEC. 12, 2057.

Angrily, Kraag burned the lettering off in a 30-second blast with his heat-gun that left the face of Stein's gravestone cherry red.

He turned to survey the terrain, and saw Jonner. The captain was crouched half a mile away, apparently writing more on a flat rock or on the ground itself.

Jonner was facing him, but his head was down and he hadn't seen Kraag. If Kraag fired the projectile pistol, he probably would miss and might warn Jonner with the shot. He was sure of his accuracy with a heat-gun. Kraag took a heat-gun in his left hand and ran toward Jonner.

Possibly the vibration of the ground warned Jonner. He looked up, jumped to his feet and fled. As soon as he could stop and get his feet planted firmly on the ground, Kraag fired the projectile pistol after him. He was still shooting low and to one side.

Kraag picked himself up from the ground, where the backlash of the weapon had knocked him, and went up to the spot where Jonner had been writing. A mathematical problem had been scratched on the surface with a sharp rock. Kraag had interrupted Jonner in the middle of it.

The figures that had been written were:

$$\begin{array}{r}
 11 \\
 \hline
 1.141 \overline{) 1552.41} \\
 \underline{141} \\
 142 \\
 \underline{141} \\
 1
 \end{array}$$

[Transcriber's note: equation in figure is long division of 1552.41 divided by 1.141]

Kraag stared at it, carrying out the rest of the simple mathematics in his head. The answer was 1101. But what was the problem?

The figure "1.41" was familiar enough. It was the square root of two, carried to two decimal places. But what was Jonner dividing by it, and why?

He frowned in concentration. There was something familiar about the numbers, something that had to do with him and Jonner, and Jonner wouldn't be working arithmetic just for amusement.

He saw Jonner moving on the horizon, just his head visible against the black sky, his body hidden by the curve of the planet. Jonner was circling.

The sudden realization of danger wiped other thoughts from his mind. Until he saw the epitaph Jonner had written for Stein, Kraag had thought Jonner looked at this as he did: one man against the other, and winner take all. But Jonner intended to win even if he lost, because Jonner was not fighting just for Jonner's survival but for due process of law.

Jonner was trying to make certain that, even if Kraag killed him, Martian law would punish Kraag for Stein's death. And if Jonner got into the sphere, he could get his message to Marsport or the rescue ship simply by turning on the radio.

Kraag turned and raced back to the sphere. He arrived, panting heavily. Jonner was nowhere in sight, but he knew Jonner, circling, could not have gotten there ahead of him.

He must kill Jonner before nightfall, if he could, but he must not get far enough from the sphere to let Jonner slip in behind him. He was not ready, yet, to destroy the radio to keep Jonner from it.

He walked around the sphere. There was Jonner on the other side, only his head above the horizon, moving clockwise. The sun flashed and gleamed from Jonner's helmet.

There was no sense in shooting at so small a target as a head. A mile away, Jonner's whole body was a small enough target. A carefully gauged leap carried Kraag to the top of the sphere. Here, 40 feet higher, his range of view was increased considerably. He could see Jonner well.

Jonner could see him, too. Jonner stopped to hurl a stone. It took a while for the missile to cover the distance. It passed below Kraag's level, some distance away from him.

"Why don't you give it up, Jonner?" asked Kraag. "You can't hurt me with a rock, at this distance."

"Why should I?" retorted Jonner. "All I have to do is wait till night."

"Sure, wait. But I'm not waiting, Jonner. One of us is going to win this thing before night, or I'm going to blast the radio so you can't reach Marsport. If I have to do that, I'll track you down tomorrow—and I think I can stay outside and fight you away from the sphere tonight."

"Getting desperate enough to fight like a man now, aren't you, Kraag? If you want a showdown today, I'm willing."

Kraag's mind was clear now. He had the situation under control. He glanced around the landscape at the scattered portions of the wrecked ship. There was the cargo hull, burst open, where Jonner had gotten his sledge hammer and the pick to bury Stein. Over there was a red sphere, ripped by the jagged

gash of the meteor collision—one of the two hydrazine fuel tanks. The yellow sphere 30 degrees away from it was an oxygen fuel tank.

Kraag leveled Jonner's gun and fired at the yellow sphere. The kick knocked him off the sphere, but as he somersaulted backwards he saw the projectile hit the ground. Still low and to one side. But he noticed something on the gun, he hadn't seen before.

There were ridges for sighting along the barrel of Jonner's pistol. Regaining his position atop the sphere, Kraag pressed his back against the observatory dome, to brace himself against the gun's backlash. He aimed carefully at the yellow sphere and fired again.



The yellow tank jumped—not from the impact, but from the spout of freed, expanding oxygen through the hole the bullet made. It moved and wobbled about in the weak gravity, like a dying balloon. When it stopped, Kraag knew he had destroyed half of Jonner's oxygen supply.

"Good shot, Kraag," congratulated Jonner, with fatalistic irony in his tone. "Of course, I'm not as big a target as the tanks."

"Each target in its own time," replied Kraag triumphantly, and looked around for the other yellow sphere.

He had been afraid it might be one of the parts that had fallen over the horizon, but it wasn't. It was behind him, a little closer than the first. He hit it with one shot.

Now Jonner had only the oxygen in his spacesuit tanks.

Jonner had made no effort to move farther away. He was still visible on the horizon, from the knees up, moving in a great circle around the personnel sphere.

Kraag aimed carefully and fired. He did not know the projectile's speed, but certainly it would be much faster than Jonner's rocks. After half a minute had passed, he knew he had missed.

There was only one thing to do. He settled himself and fired again, trying to lead Jonner slightly. Again he missed.

Methodically, taking his time, Kraag fired. Jonner walked on unconcernedly, circling. Kraag tried to fire so the path of his projectile would strike at the top of Jonner's strides, for then Jonner rose several feet into the air and his whole body was visible.

Occasionally, Jonner would stop and hurl a stone at Kraag. One man was as inaccurate as the other. Jonner's stones went wide at that distance, and Kraag obviously had not hit Jonner with a bullet.

At last Jonner stopped. He seemed to be fiddling with something that was right on the ground, below Kraag's line of vision. Then a tremendous stone, bigger than Kraag's head, came hurtling toward the sphere. Kraag ducked instinctively, but the missile passed 10 feet above him, still going well.

"What in the devil!" exclaimed Kraag.

"A little innovation of mine, to make things more interesting," said Jonner. "In case you ever want to use the idea, I made me a super-slingshot out of two of the jeep inner tubes from the cargo, and a couple of crowbars I could drive into crevices. Fixed it up yesterday for bombardment purposes."

The duel went on.

There came the time when the hammer of the pistol clicked on an empty chamber.

"How do you refuel this thing, Jonner?" asked Kraag pleasantly. The sun was still high. He could retreat to the interior of the sphere and figure it out if he had to.

"It's pretty hard to do with spacesuit hooks," replied Jonner. "Be glad to demonstrate, if you'll toss me the gun."

Kraag laughed, a laugh with more triumph in it than humor, because in his fumbling he had just hit the button that ejected the magazine. To push in a fresh one was the matter of a moment.

He had hoped Jonner would move in closer when he knew the pistol was empty, but no such luck. Jonner stayed put.

Kraag's first effort with the new magazine brought no results, for he had neglected to prime the weapon by pushing the outer covering back on the barrel. He did this, and resumed his methodical firing.

As the time wore on, Kraag began to appreciate the difficulties involved in hitting a moving target, even a slowly moving one, when the marksman was as inexperienced as he was. The trouble was that, at that distance, he could not see where the bullets were striking and had no way of knowing how wide of his mark he was shooting.

He was on the fourth magazine and the sun had passed the meridian when he felt the sphere vibrate faintly and momentarily beneath him. He twisted around, alarmed. He could see nothing. It wasn't one of Jonner's rocks, because a big one had just missed.

His eye detected a shining streak that stretched a few inches along the curve of the sphere's meteor shield, at about the level of his feet. He bent to examine it. Something had struck it at high speed, a glancing blow.

It couldn't be one of Jonner's rocks. Small meteor?

A jagged hole suddenly appeared in the observatory dome near him. Kraag moved up and examined it closely. It had been made by some small object. Through the glassite he could see a similar hole in the other side of the dome.

Did Jonner have some sort of new weapon? He couldn't. Even Jonner wasn't resourceful enough to invent a high-powered weapon with the innocuous cargo they were carrying for the Titan colony.

Something struck Kraag a powerful blow in the left chest, a blow that hurled him sideways, to tumble off the sphere and fall slowly to the ground below. There was a great pain in his chest, and he released his hand-hooks in agony, so that the pistol fell away from him.

Kraag gasped for breath as he struck the ground and bounced. He coughed up blood.

He fell slowly again, and bounced again. The third time he settled jarringly, prone on his back.

He couldn't understand what had happened to him. He pulled his right arm inside the suit with an effort and probed the painful area on his chest. He felt the hot wetness of flowing blood.

He would have to get to the sphere. He tried to move. He couldn't get off his back. He lay there and writhed in pain.

Jonner's voice was in his ears, saying something.

"I knew it would get you," Jonner said. "It was my only chance. But it got you at last, Kraag."

"Come help me, Jonner," whimpered Kraag weakly. "I've been hit by ... I don't know. It must have been a meteor."

"I'm coming as fast as I can, Kraag, but it was no meteor. It was my gun."

"Gun?" repeated Kraag wonderingly.

"I warned you about that gun of mine, Kraag. If you'd looked at the figures on the barrel, the muzzle velocity of those .45-calibre bullets is 1100 feet a second. With Ceres' escape velocity, that's almost exactly the circular velocity at the asteroid's surface."

Jonner was standing over him, and then was lifting him gently, to carry him to the sphere.

"I deliberately got just out of your range of vision, from the ground, so you'd climb to a high spot," said Jonner. "You had to be high, so the bullet would clear the irregularities on the planet's surface, and I knew that sooner or later you'd shoot a bullet or two high enough not to hit the ground."

"When you were firing at me, your bullets weren't describing a trajectory and falling to the surface, as they would on Earth or Mars. They were taking

an orbit that brought them all the way around the planet to the same spot, to hit you from the other side two hours later."

Kraag tried to look up at him. Something was going wrong with his sight, and everything outside his face plate was a blur. Must be the oxygen ... maybe his suit didn't seal the bullet hole properly.

"I thought...." Kraag began, and choked. He coughed, slowly and painfully, then tried again: "I thought that ... problem on the rocks ... looked familiar."

"You've always done it with a slide rule. That's probably why the long division didn't register," said Jonner. "The equation is one every spaceman knows: the circular velocity equals the escape velocity divided by the square root of two."

*** END OF THE PROJECT GUTENBERG EBOOK ESCAPE
VELOCITY ***

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